

Image Enhancement and Preprocessing continued

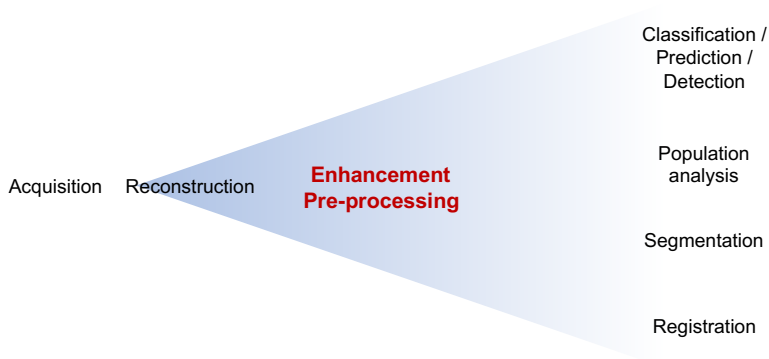
Ender Konukoglu

ETH Zürich

February 24 - March 3, 2026

Section 1

Introduction



Enhancement and pre-processing tasks form initial steps for any type of analysis. Same applies to methods that use machine learning, probabilistic, variational or energy-based formulations.

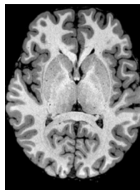
Image enhancement and pre-processing

Enhancement of four conditions

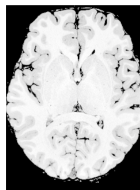
- Bad contrast
- Varying intensity statistics
- Noise
- Bias field
- Variation in pixel-size

Why enhancement?

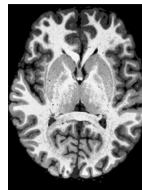
- simplifying interpretation
- better visualization
- normalization for further processing



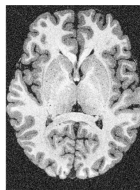
Original Image



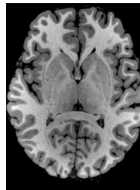
Bad contrast



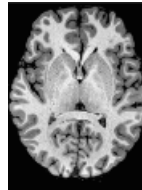
Intensity variation



Noise



Bias field



Different pixel-size

Outline

- Intensity normalization
- Noise suppression
- Exercise / Challenge
- Bias correction
- Variations in image and pixel-size

Section 3

Noise suppression

Subsection 4

Energy-based

Optimization-based methods

Remember the additive model for the noisy image.

$$\underbrace{J(x)}_{\text{Observed}} = \underbrace{I(x)}_{\text{Desired}} + \underbrace{\epsilon(x)}_{\text{Noise}}$$

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Denoising as an energy minimization problem

$$\hat{I} = \arg_I \min \lambda D(I, J) + R(I)$$

- $D(I, J)$: Data consistency
- $R(I)$: Regularization - embedding prior information (opinions) on the desired image.
- λ : Weighting coefficient.

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Alternative constrained optimization

$$\hat{I} = \arg_I \min R(I), \text{ such that } D(I, J) = 0$$

Total variation denoising

[Rudin, Osher and Fatemi, Physica D, 1992]

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- Data consistency:

$$D(I, J) = \int (I(x) - J(x))^2 dx = \|I - J\|_2^2$$

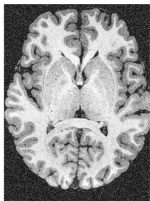
- Regularization:

$$R(I) = \int \|\nabla I\| dx = \|I\|_{TV}$$

- λ : Allowed noise level in the observed image with $\lambda = 1/2\sigma^2$

Total variation denoising: Experiments

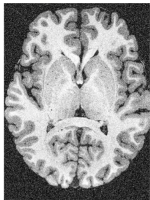
Total variation aims to get an image with minimal variation.



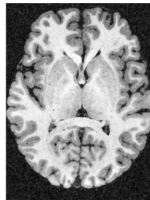
noisy image

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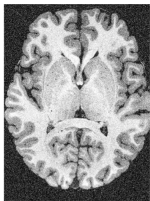
noisy image



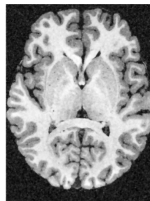
$\lambda = 5.0$

Total variation denoising: Experiments

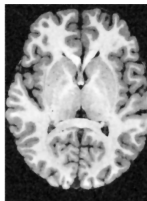
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noisy image



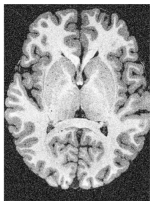
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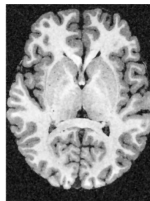
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Total variation denoising: Experiments

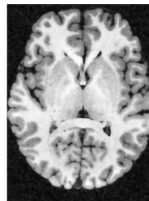
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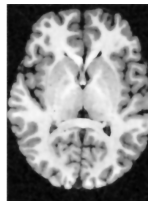
noisy image



$\lambda = 5.0$



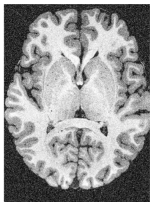
$\lambda = 3.5$



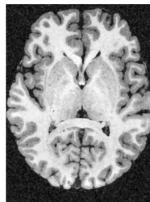
$\lambda = 2$

Total variation denoising: Experiments

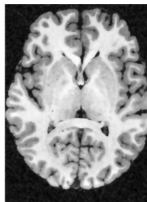
Total variation aims to get an image with minimal variation.



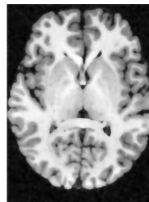
noisy image



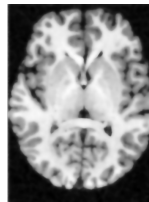
$\lambda = 5.0$



$\lambda = 3.5$



$\lambda = 2$



$\lambda = 1.0$

Probabilistic interpretation

The energy-based formulation can be interpreted in a probabilistic model. Remember that

$$\arg_I \min \lambda D(I, J) + R(I) = \arg_I \max \log P(J|I) + \log P(I)$$

- $P(J|I)$: Likelihood
- $P(I)$: Prior

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Total variation denoising model:

$$\begin{aligned} P(J|I) &= \mathcal{N}\left(J; I, \frac{1}{2\lambda}\right) \\ P(I) &\propto \exp(-\|\nabla I\|) \end{aligned}$$

Remark

Note that the standard deviation of the likelihood model is linked to the weight in the energy-based formulation.

Generality

Note that the energy-based or the corresponding probabilistic formulations are very generic and many different models can be cast in the same formulation.

$$\arg_I \min \lambda D(I, J) + R(I) = \arg_I \max \log P(J|I) + \log P(I)$$

- $D(I, J)$ or $P(J|I)$ encode information how the noise process works. It does not have to be additive Gaussian noise but it can be different things as we have seen before.
- $R(I)$ or $P(I)$ encode our prior information on what type of images are plausible.
 - TV norm: minimal variation
 - Wavelet: sparse in a transform domain
 - Dictionary learning: composed of sparse atomic elements.
 - Unsupervised learning: probabilistic models learned from data.
 - Adversarial learning: comparing distribution wise.

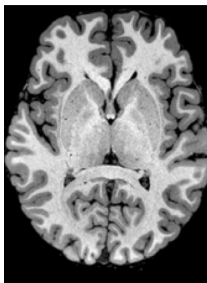
Section 4

Bias Correction

Outline

- Intensity normalization
- Noise suppression
- Exercise / Challenge
- Bias correction
 - What is bias field?
 - N3 algorithm
- Variations in image and pixel-size

What is bias field?



Normal image

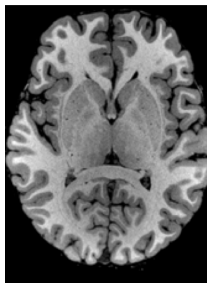
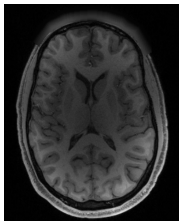


Image with bias

- Acquisition artifact
 - Field inhomogeneity
 - Electrical characteristics of the tissue
 - Poor uniformity of coils
 - Gradients and eddy currents
- Most MRI has it
- Minimal effect on visual interpretation. Actually difficult to see by eye
- Adverse affects on algorithms, esp. segmentation
- Non-linear effect on intensities that vary spatially
- Remove it as a pre-processing step

What does it look like?



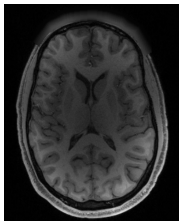
Image



Bias field

- Smoothly varying
- Does not depend on the underlying tissue to the first order approximation
- In reality it also depends on the tissue but this effect is mostly ignored

What does it look like?



Image



Bias field

- Smoothly varying
- Does not depend on the underlying tissue to the first order approximation
- In reality it also depends on the tissue but this effect is mostly ignored
- Approximate mathematical model:

$$j = bi + n,$$

i : real image,

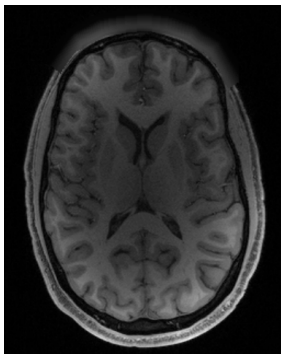
b : bias field,

n : noise and

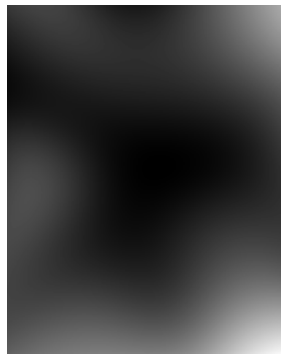
j : observed image

- Multiplicative factor

How would you do it?

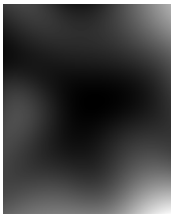
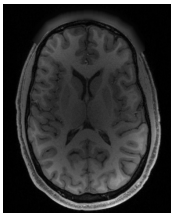


Image

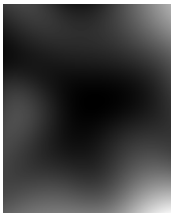
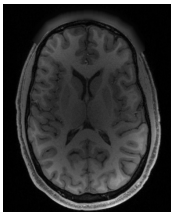


Bias field

Basic methods to correct it

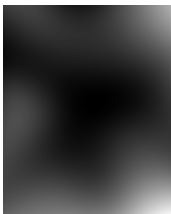
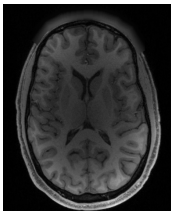


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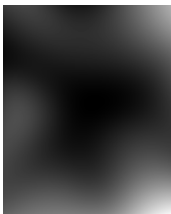
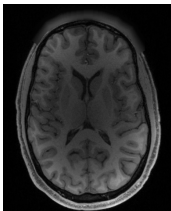
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 - Ideally a perfect solution
 - Additional measurements makes it hard for retrospective data analysis and usually not integrated
 - Post-acquisition methods are very useful

Basic methods to correct it



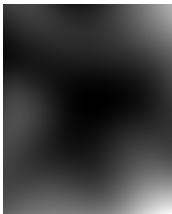
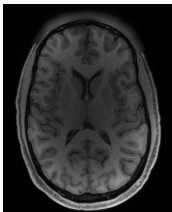
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 - If you know what you expect in the image
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- N3 algorithm
 - Generic
 - Most popular solution
 - [Sled, Zijdebos and Evans, IEEE TMI 1998]
 - We will study N3 in this lecture.

N3 algorithm - principle

Starting from the main model

$$\tilde{j}(x) = \tilde{b}(x)\tilde{i}(x) + \tilde{n}(x)$$

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Goal

N3 estimates $b(x)$ at every point in the image. Bias removed image can be immediately computed by $i(x) = j(x) - b(x)$.

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Assumptions

(1) Independence between pixels when considering distributions of j , b and i . (2) Bias field and the image intensities are independent.

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Assumptions

(1) Independence between pixels when considering distributions of j , b and i . (2) Bias field and the image intensities are independent.

$j(x) = b(x) + i(x)$ is a sum of random variables with pdf denoted as $J(j)$, $B(b)$ and $I(i)$. Let's study how distribution of j is related to the distribution of i and b .

Sums of random variables

Let a and b be two independent random variables with pdfs denoted by $f_a(a)$ and $f_b(b)$. Let us define another random variable

$$c = a + b$$

Can we compute the distribution of c , i.e. $f_c(c)$, using distributions of a and b ?

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Sums of random variables

The CDF is given as

$$F_c(c') = \int_{-\infty}^{\infty} \int_{\infty}^{c'-b} f_a(a) f_b(b) da db.$$

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Using Leibnitz' rule twice:

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Convolution

$c = a + b \Rightarrow f_c = f_a * f_b$ when a and b are independent random variables.

Back to bias problem

$$j(x) = b(x) + i(x),$$

Denote the probability distributions of j , b and i with $J(j)$, $B(b)$ and $I(i)$. The probability distribution J is the following convolution:

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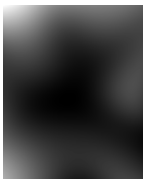
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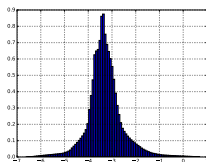
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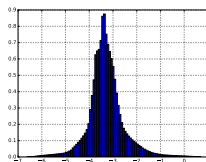
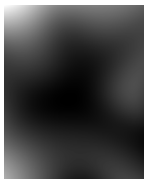


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What can we say about B ?

- $\tilde{j}(x) = \tilde{i}(x)\tilde{b}(x)$
- $\tilde{b}$ is around 1 so b is around 0
- B can be approximated with a Gaussian with small standard deviation
- J is a smoothed version of I
- The bias field is a smooth field - b is smooth

Two components in the problem

$$j(x) = b(x) + i(x),$$

Denote the probability distributions of j , b and i with $J(j)$, $B(b)$ and $I(i)$. The probability distribution J is the following convolution:

$$J(j) = \int I(j - b)B(b)db = I * B$$

The estimation must determine $I(i)$ and b given only j .

Two components in the problem

$$j(x) = b(x) + i(x),$$

Denote the probability distributions of j , b and i with $J(j)$, $B(b)$ and $I(i)$. The probability distribution J is the following convolution:

$$J(j) = \int I(j - b)B(b)db = I * B$$

The estimation must determine $I(i)$ and b given only j .

We divide this into two parts:

- Field estimation: Estimation of b given $I(i)$
(note $I(i)$ is only the distribution and not the image, and we are looking for b).
- $I(i)$ estimation: Estimation of $I(i)$.

Always assuming B is a Gaussian with small standard deviation.

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 \end{aligned}$$

We know all the terms in this equation and i is one dimensional, so, we can solve it numerically with ease.

Component 2: $I(i)$ estimation

Key observation

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with Z being a small number and $\mathcal{F}\{B\}^*$ denotes the complex conjugate.

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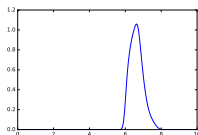
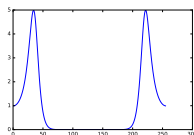
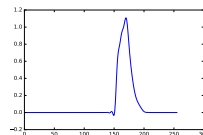
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Note: This is inverse filtering!

Intermediate summary

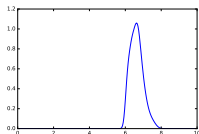
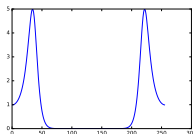
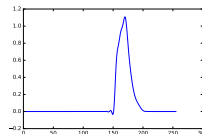
- Estimate of the unbiased image's distribution: $I(i) \approx \mathcal{F}^{-1} \left(\hat{\mathcal{F}}\{J\} \right)$


 $J(j)$

 $|\hat{\mathcal{F}}|$

 $I \approx \mathcal{F}^{-1}(\hat{\mathcal{F}}\{J\})$

- field estimate: $b_e(j) = j - \frac{\int_{-\infty}^{\infty} iB(j-i)I(i)di}{\int_{-\infty}^{\infty} B(j-i)I(i)di}$
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with K smoothing kernel such as $\exp(\|x - x_n\|_2^2 / \sigma^2)$.

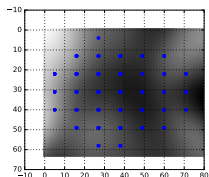
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- Example



2. Need for iterations

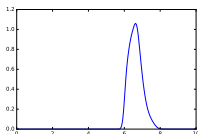
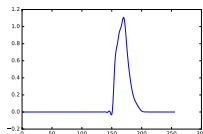
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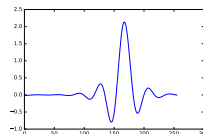
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 $J(j)$ 

std = 0.1

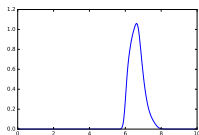
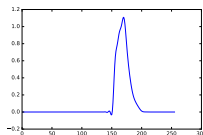


with std = 0.5

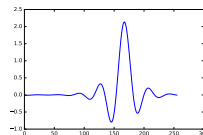
Remember they have to be distributions (> 0). Deconvolution is hard.

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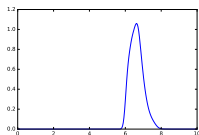
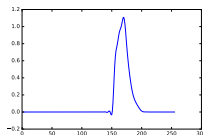
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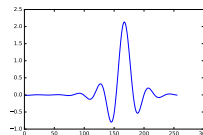
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with std = 0.5

Remember they have to be distributions (> 0). **Deconvolution is hard.**

- Solution: Iterate many times with small Gaussians (small stds)
- Not a bad approximation: large Gaussians are convolutions of smaller ones

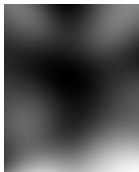
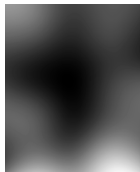
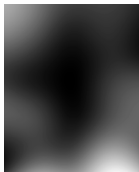
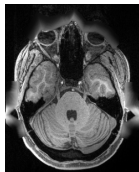
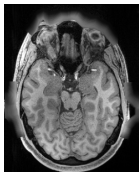
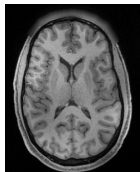
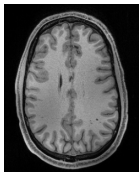
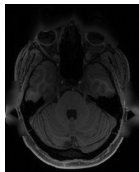
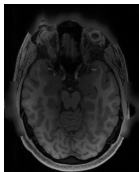
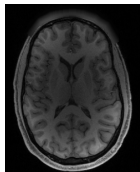
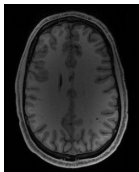
Algorithm

Algorithm 1 Bias correction with N3

- 1: $b = 0, j = \log(\tilde{j})$ observed image
 - 2: $\hat{F} = \frac{\hat{B}^*}{|\hat{B}|^2 + Z^2}$
 - 3: **while** $\|\Delta b\| \geq \text{eps}$ **do**
 - 4: Estimate $J(j)$ with kernel density estimation or simple histogram
 - 5: Compute $I(i) = \mathcal{F}^{-1}(\hat{F}\mathcal{F}\{J\})$
 - 6: Compute $\Delta b_e = j - \mathbb{E}(i|j)$ at x_n anchor points.
 - 7: Get the smooth estimate $\Delta b(x) = \sum_{n=1}^N \Delta b_e(j(x_n))K(\|x - x_n\|)$ for all voxels
 - 8: Update $b = b + \Delta b$
 - 9: Update $j = j - \Delta b$
 - 10: **end while**
-

- └ Bias Correction
- └ N3 algorithm

Examples



Analysis

- Bias correction with N3 is routinely used for MRI today
- There are variations available, e.g. N4ITK
- In the method there are multiple approximations
- Multiresolution solutions exist and perform better
- May need to run the tool a few times to get better results

Section 5

Variations in image and pixel-size

Outline

- Contrast enhancement
- Intensity normalization
- Noise suppression
- Exercise / Challenge
- Bias correction
- Variations in image and pixel-size
 - What is the problem?
 - How to tackle it.
 - How **not** to tackle it.

Different image and pixel size and FOV



Fine pixel-size, small FOV



Coarse pixel-size, large FOV

What is it?

- Images may come with different pixel and/or image size.
- They may have different Field-of-View (FOV).
- You may need to compare such “heterogeneous” images.

Why does it happen?

- Differences in acquisitions
- in FOV
- in pixel size

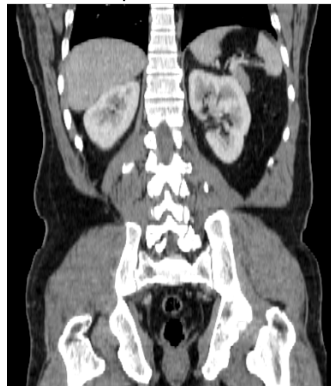
- └ Variations in image and pixel-size
- └ What is the problem?

Why is this a problem?

Sizes of the images on the slides are proportional to their number of pixels.



Fine pixel-size, small FOV



Coarse pixel-size, large FOV

In which applications does it matter?

- Comparing measurements
- Registration
- Machine learning applications

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 - measurements taken in an image have a correspondence in real life.
 - real life measurements rely on pixel size.
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- Effectively important for all applications
- Building algorithms that work for heterogeneous images is one of the most challenging tasks

How do we tackle this?

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

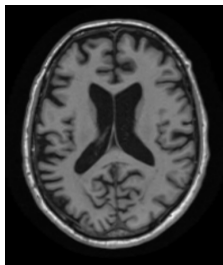


Image size: [454, 384]
Pixel size: [1.25, 1.25]

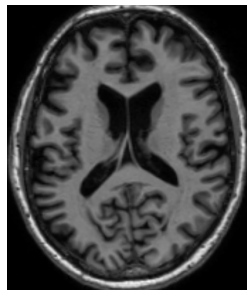


Image size: [494, 424]
Pixel size: [1.0, 1.0]

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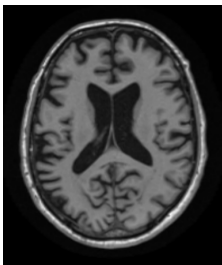


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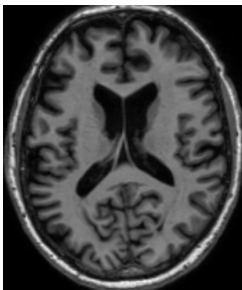


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Measurements taken in the two images should be comparable.

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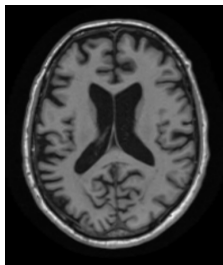


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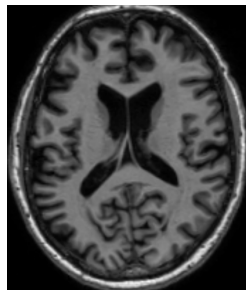


Image size: [494, 424]
Pixel size: [1.0, 1.0]

- What criteria should guide us?
Measurements taken in the two images should be comparable.
- What is the important factor to match?

How do we tackle this?

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

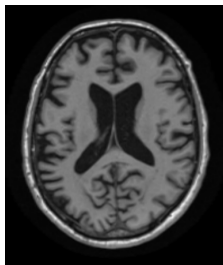


Image size: [454, 384]
Pixel size: [1.25, 1.25]

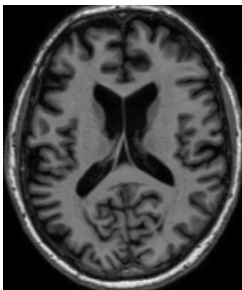


Image size: [494, 424]
Pixel size: [1.0, 1.0]

- What criteria should guide us?
Measurements taken in the two images should be comparable.
- What is the important factor to match? **Pixel size**

Match the pixel size

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

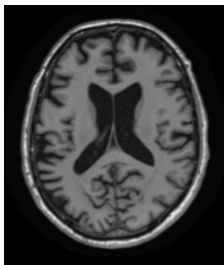


Image size: [454, 384]

Pixel size: [1.25, 1.25]

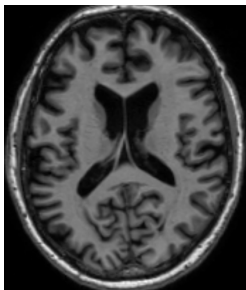


Image size: [494, 424]

Pixel size: [1.0, 1.0]

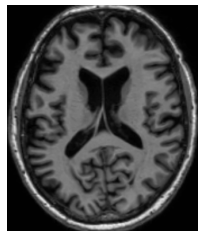


Image size: [395, 339]

Pixel size: [1.25, 1.25]

- Resample one image to match the *pixel-size* of the other.
- Resize an image with a factor determined by the pixel-size. One example tool in python “`scipy.ndimage.zoom`”.

Match the pixel size

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

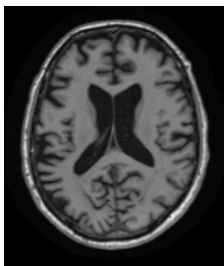


Image size: [454, 384]
Pixel size: [1.25, 1.25]

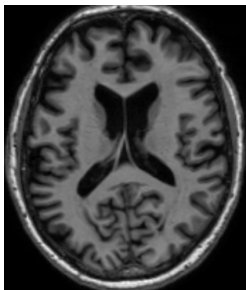


Image size: [494, 424]
Pixel size: [1.0, 1.0]

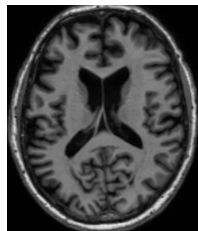


Image size: [395, 339]
Pixel size: [1.25, 1.25]

- Resample one image to match the *pixel-size* of the other.
- Resize an image with a factor determined by the pixel-size. One example tool in python “`scipy.ndimage.zoom`”.
- If volumetric images in 3D, then match the *voxel-size*.

Match the pixel size

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

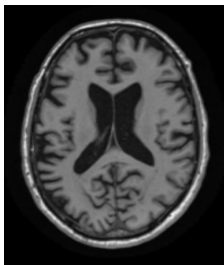


Image size: [454, 384]
Pixel size: [1.25, 1.25]

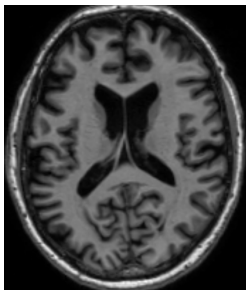


Image size: [494, 424]
Pixel size: [1.0, 1.0]

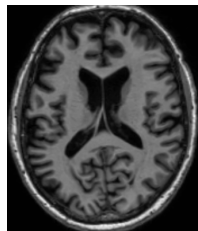


Image size: [395, 339]
Pixel size: [1.25, 1.25]

- Resample one image to match the *pixel-size* of the other.
- Resize an image with a factor determined by the pixel-size. One example tool in python “`scipy.ndimage.zoom`”.
- If volumetric images in 3D, then match the *voxel-size*.
- Crop or pad if necessary to make the image sizes the same.

- └ Variations in image and pixel-size
- └ How to tackle it.

Exercise

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

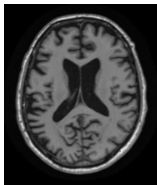


Image size: [454, 384]
Pixel size: [1.25, 1.25]

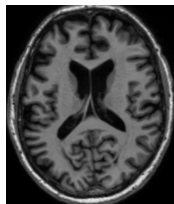


Image size: [494, 424]
Pixel size: [1.0, 1.0]

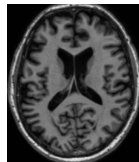


Image size: [395, 339]
Pixel size: [1.25, 1.25]

- If we are matching the image in the second column to the one in the first one, what are the resize ratios in x- and y-directions?

- └ Variations in image and pixel-size
- └ How to tackle it.

Exercise

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

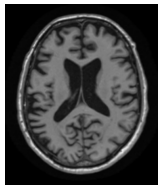


Image size: [454, 384]
Pixel size: [1.25, 1.25]

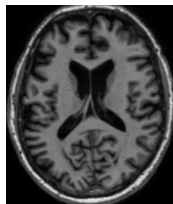


Image size: [494, 424]
Pixel size: [1.0, 1.0]

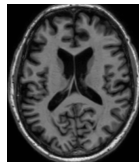


Image size: [395, 339]
Pixel size: [1.25, 1.25]

- If we are matching the image in the second column to the one in the first one, what are the resize ratios in x- and y-directions? **0.8, 0.8**

Exercise

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

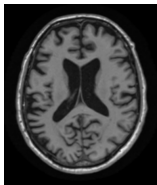


Image size: [454, 384]
Pixel size: [1.25, 1.25]

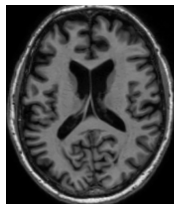


Image size: [494, 424]
Pixel size: [1.0, 1.0]

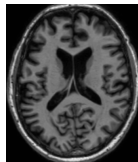


Image size: [395, 339]
Pixel size: [1.25, 1.25]

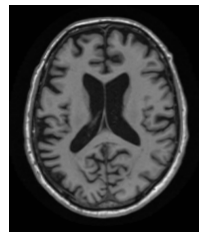


Image size: [568, 480]
Pixel size: [1.0, 1.0]

- If we are matching the image in the second column to the one in the first one, what are the resize ratios in x- and y-directions? **0.8, 0.8**
- If we are matching the image in the first column to the one in the second one, what are the resize ratios in x- and y-directions?

Exercise

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

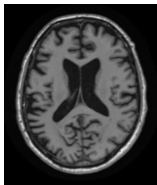


Image size: [454, 384]
Pixel size: [1.25, 1.25]

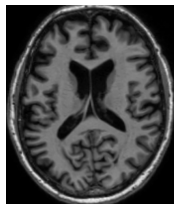


Image size: [494, 424]
Pixel size: [1.0, 1.0]

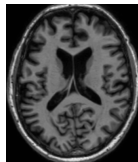


Image size: [395, 339]
Pixel size: [1.25, 1.25]

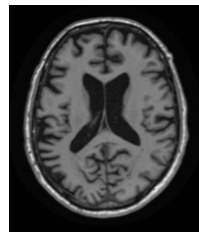


Image size: [568, 480]
Pixel size: [1.0, 1.0]

- If we are matching the image in the second column to the one in the first one, what are the resize ratios in x- and y-directions? **0.8, 0.8**
- If we are matching the image in the first column to the one in the second one, what are the resize ratios in x- and y-directions? **1.25, 1.25**

Where do I find this information?

- We need the pixel-size of each image for the matching.
- **Such information are stored in the header of the DICOM files.**
- It is also stored in the headers of many volumetric medical image formats.
- One way to reach them is to use the “nibabel” package in python.

How **not** to tackle it.

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

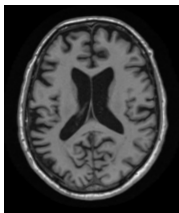


Image size: [454, 384]
Pixel size: [1.25, 1.25]

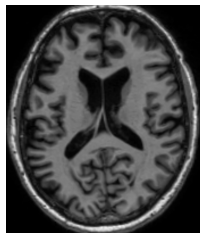


Image size: [494, 424]
Pixel size: [1.0, 1.0]

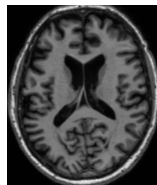


Image size: [395, 339]
Pixel size: [1.25, 1.25]

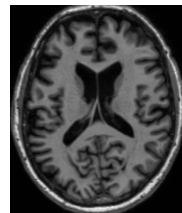


Image size: [454, 384]
Pixel size: [1.09, 1.11]

- A very naive way of doing the matching is through matching the image sizes.
- Last column on the right is obtained by matching the size of the image on the second column to that in the first one.
- Notice that the images are not comparable any more.
- Even the aspect ratio changed.

How **not** to tackle it.

Images of the same person one year apart

Sizes of the images on the slides are proportional to their number of pixels.

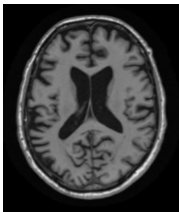


Image size: [454, 384]
Pixel size: [1.25, 1.25]

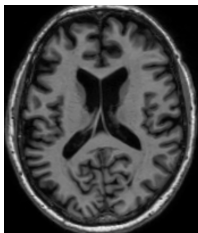


Image size: [494, 424]
Pixel size: [1.0, 1.0]

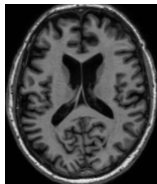


Image size: [395, 339]
Pixel size: [1.25, 1.25]

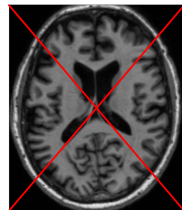


Image size: [454, 384]
Pixel size: [1.09, 1.11]

- A very naive way of doing the matching is through matching the image sizes.
- Last column on the right is obtained by matching the size of the image on the second column to that in the first one.
- Notice that the images are not comparable any more.
- Even the aspect ratio changed.

Important

Always match the pixel / voxel-size. Do not match the image size, it may lead to wrong normalization.